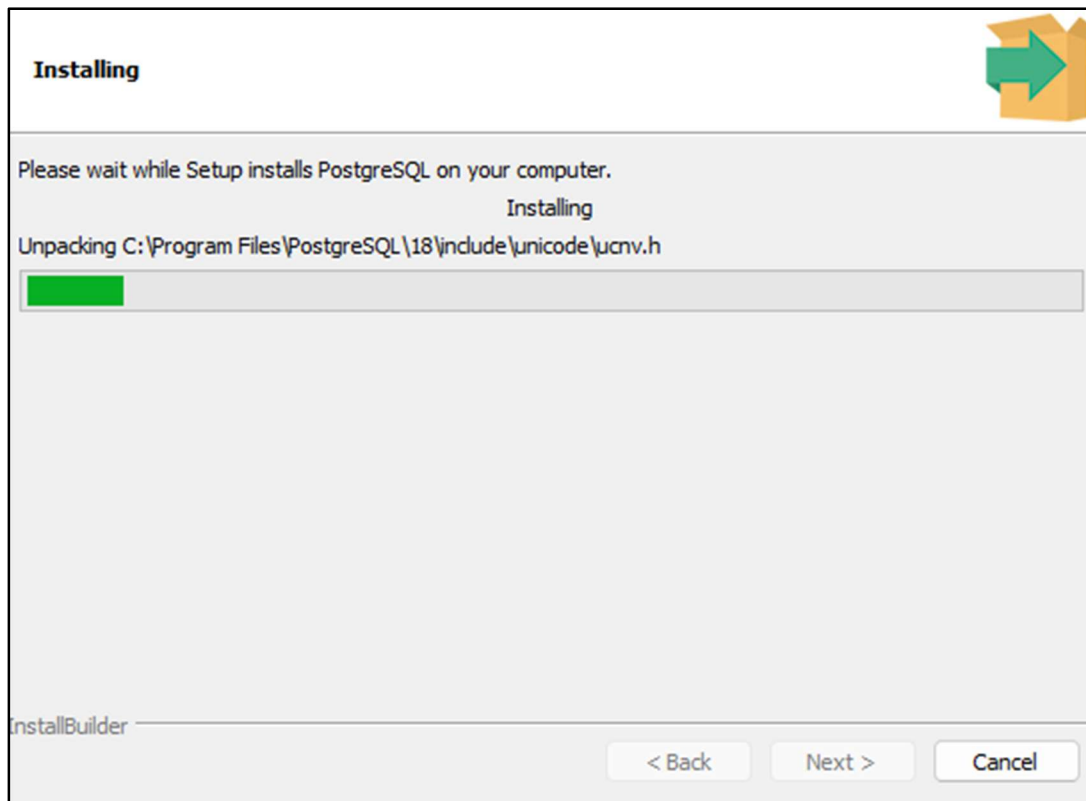


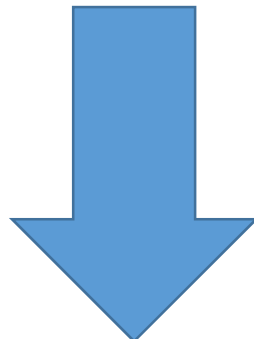
Evidencia de instalación de PSQL:



Comprobación de PSQL y creación de base de datos:

```
Server [localhost]: 127.0.0.1
Database [postgres]: postgres
Port [5432]: 5432
Username [postgres]: postgres
Password for user postgres:
psql (18.6)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=#
```



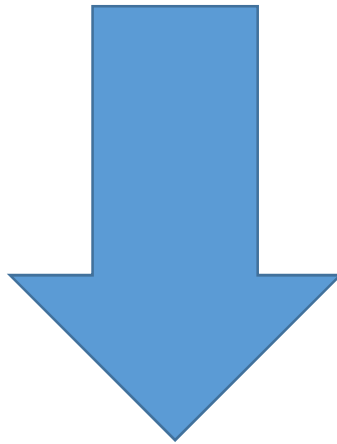
```

postgres=# CREATE DATABASE campus_tasks;
CREATE DATABASE
postgres=# \c campus_tasks
You are now connected to database "campus_tasks" as user "postgres".
campus_tasks=# CREATE TABLE tareas (id SERIAL PRIMARY KEY, titulo TEXT NOT NULL);
CREATE TABLE
campus_tasks=# INSERT INTO tareas (titulo) VALUES
campus_tasks-# ('Leer la guía de la clase 2'),
campus_tasks-# ('Preparar el entorno de desarrollo');
INSERT 0 2
campus_tasks=# SELECT * FROM tareas
campus_tasks-# SELECT * FROM tareas;
ERROR:  syntax error at or near "SELECT"
LINE 2: SELECT * FROM tareas;
        ^
campus_tasks=# SELECT * FROM tareas;
 id |          titulo
----+-----
  1 | Leer la guía de la clase 2
  2 | Preparar el entorno de desarrollo
(2 rows)

campus_tasks=#

```

Comprobación del git pull origin Master solicitado (migrar cambios a rama local):



```
backend > src > database > TS database.service.ts > ...
1  import { Injectable, OnModuleDestroy } from '@nestjs/common';
2  import { Pool, QueryResult, QueryResultRow } from 'pg';
3
4  @Injectable()
5  export class DatabaseService implements OnModuleDestroy {
6    private readonly pool = new Pool({
7      host: process.env.DB_HOST,
8      port: Number(process.env.DB_PORT ?? 5432),
9      user: process.env.DB_USER,
10     password: process.env.DB_PASSWORD || undefined,
11     database: process.env.DB_NAME,
12   });
13
14   query<T extends QueryResultRow>({
15     sql: string,
16     params: unknown[] = [],
17   }): Promise<QueryResult<T>> {
18     return this.pool.query<T>(sql, params);
19   }
20 }
```

Problems 2 Output Debug Console Terminal Ports

```
PS C:\Users\Gabriel\Documents\GitHub\campus-task> git pull origin master
create mode 100644 .idea/misc.xml
create mode 100644 .idea/modules.xml
create mode 100644 .idea/vcs.xml
create mode 100644 backend/src/database/database.module.ts
create mode 100644 backend/src/database/database.service.ts
create mode 100644 backend/src/tareas/tareas.integration.spec.ts
create mode 100644 backend/src/tareas/tareas.service.spec.ts
create mode 100644 frontend/src/components/tareas/tareas.component.spec.ts
PS C:\Users\Gabriel\Documents\GitHub\campus-task> 
```

Comprobación de backend:

```
Problems 1 Output Debug Console Terminal Ports

[1:32:30 PM] Found 0 errors. Watching for file changes.

[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [NestFactory] Starting Nest application...
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [InstanceLoader] AppModule dependencies initialized +25ms
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [InstanceLoader] DatabaseModule dependencies initialized +0ms
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [InstanceLoader] TareasModule dependencies initialized +1ms
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [RoutesResolver] TareasController {/tareas}: +8ms
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [RouterExplorer] Mapped {/tareas, GET} route +6ms
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [RouterExplorer] Mapped {/tareas, POST} route +3ms
[Nest] 139888 - 10/01/2026, 1:32:32 PM LOG [NestApplication] Nest application successfully started +10ms

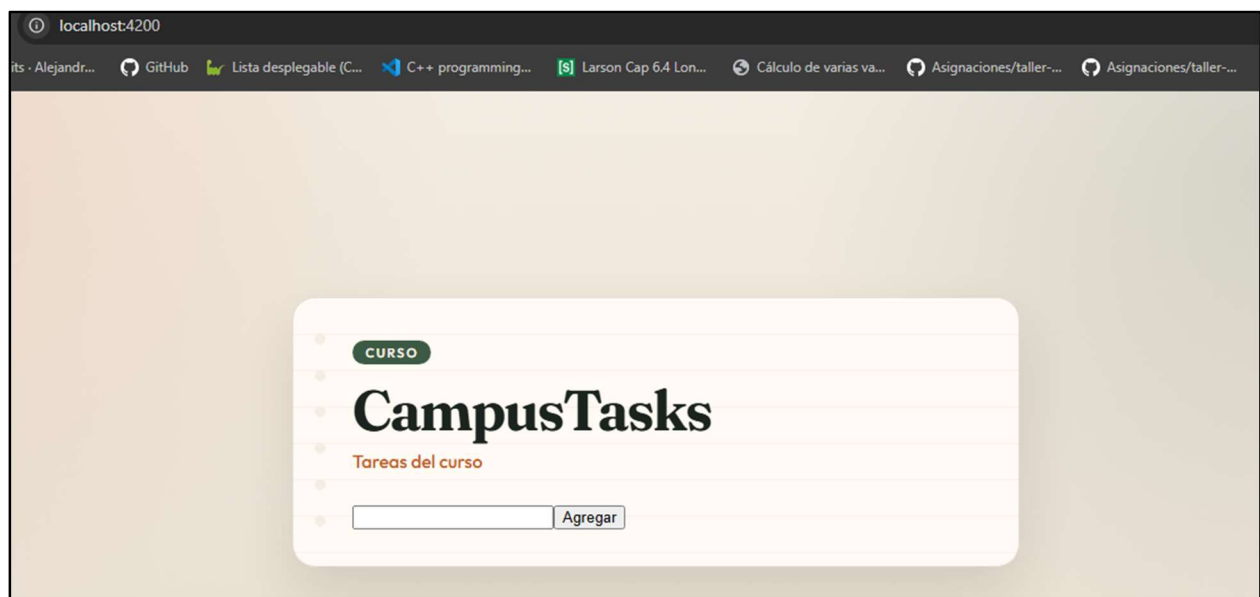
```

Comprobación de localhost:3000 en backend:

```
Impresión con formato estilístico ☒

[
  {
    "id": 1,
    "titulo": "Leer la guía de la clase 2"
  },
  {
    "id": 2,
    "titulo": "Preparar el entorno de desarrollo"
  }
]
```

Comprobación de frontend al inicializar el localhost:4200 en el navegador:



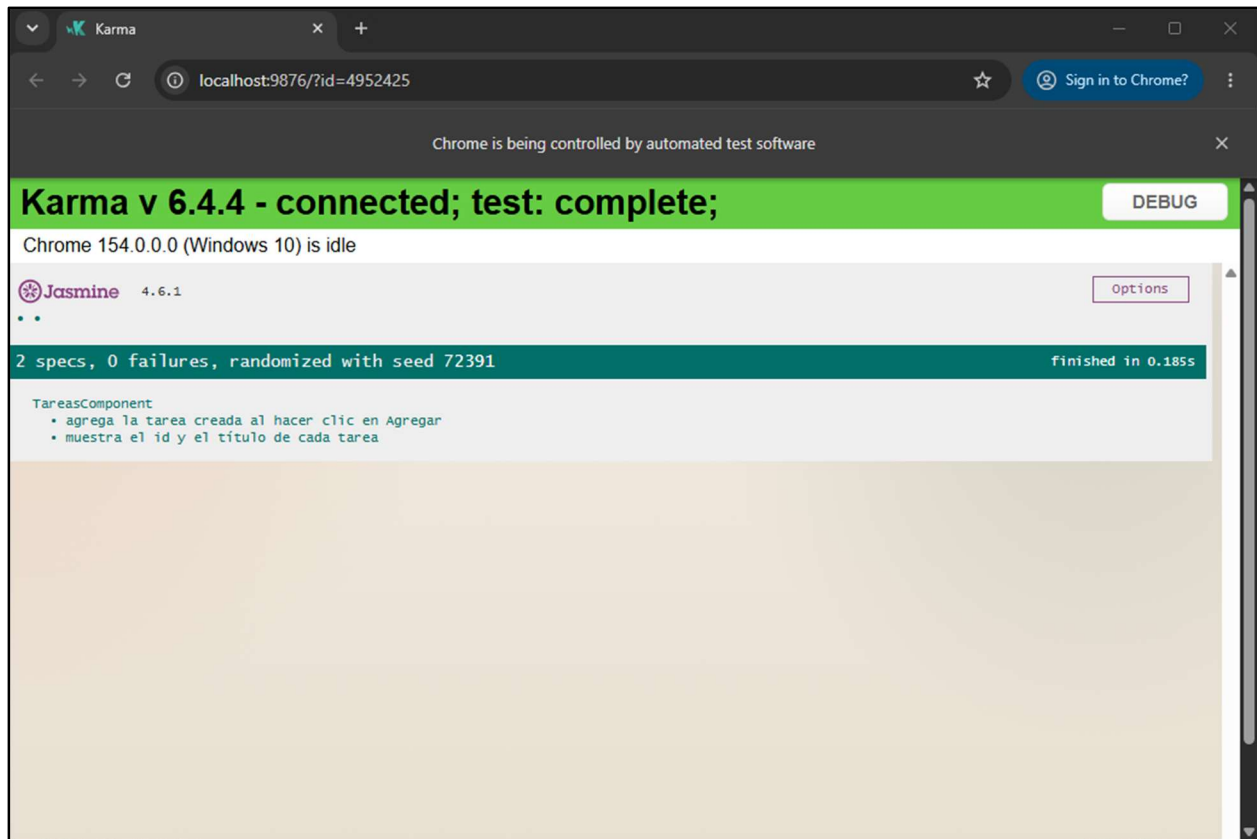
Comprobacion de tests y suites en backend:

```
PS C:\Users\Gabriel\Documents\GitHub\campus-task\backend> npm test
> jest

PASS src/tareas/tareas.service.spec.ts
PASS src/tareas/tareas.integration.spec.ts

Test Suites: 2 passed, 2 total
Tests:       4 passed, 4 total
Snapshots:  0 total
Time:        3.618 s
Ran all test suites.
PS C:\Users\Gabriel\Documents\GitHub\campus-task\backend> 
```

Comprobación de test en frontend (Karma):



Comprobacion de backend con modificaciones realizadas usando npm test:

```
PASS src/tareas/tareas.integration.spec.ts
PASS src/tareas/tareas.service.spec.ts

Test Suites: 2 passed, 2 total
Tests:       10 passed, 10 total
Snapshots:   0 total
Time:        2.529 s
Ran all test suites.
PS C:\Users\Gabriel\Documents\GitHub\campus-task\backend>
```

Comprobación del frontend en localhost:4200 con modificaciones realizadas (Agregar y Editar):



Comprobación del backend en localhost:3000 con modificaciones realizadas (Agregado y Editado):

Impresión con formato estilístico ☒

```
[
  {
    "id": 1,
    "titulo": "Leer la guía de la clase 2"
  },
  {
    "id": 2,
    "titulo": "Preparar el entorno de desarrollo"
  },
  {
    "id": 3,
    "titulo": "Tarea de prueba (editado)"
  }
]
```

Impresión con formato estilístico ☒

```
[
  {
    "id": 1,
    "titulo": "Leer la guía de la clase 2"
  },
  {
    "id": 2,
    "titulo": "Preparar el entorno de desarrollo"
  },
  {
    "id": 3,
    "titulo": "Tarea de prueba"
  }
]
```

Comprobación del PSQL con cambios realizados en el backend:

```
postgres=# \c campus_tasks
You are now connected to database "campus_tasks" as user "postgres".
campus_tasks=# SELECT * FROM tareas;
 id |
-----+-----
  1 | Leer la guía de la clase 2
  2 | Preparar el entorno de desarrollo
  3 | Tarea de prueba (editado)
(3 rows)

campus_tasks=#
```

Comprobación de Karma con modificaciones realizadas:

Chrome is being controlled by automated test software

Karma v 6.4.4 - connected; test: complete;

DEBUG

Chrome 154.0.0.0 (Windows 10) is idle

 **Jasmine** 4.6.1

Options

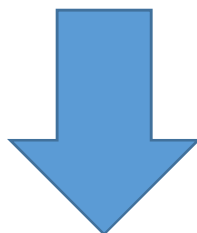
4 specs, 0 failures, randomized with seed 89167

finished in 0.195s

TareasComponent

- edita una tarea tras pulsar Editar, cambiar el texto y pulsar Guardar
- elimina una tarea de la lista tras pulsar Eliminar
- muestra el id y el título de cada tarea
- agrega la tarea creada al hacer clic en Agregar

Comprobación del frontend mediante npm test con modificaciones realizadas:



```
PS C:\Users\Gabriel\Documents\GitHub\campus-task\frontend> npm test
```

spec-components-tareas-tareas.component.spec.js	spec-components-tareas-tareas.component.spec	111.32 kB
jasmine-cleanup-1.js	jasmine-cleanup-1	67.17 kB
chunk-30QY4ED3.js	-	7.40 kB
styles.css	styles	572 bytes
jasmine-cleanup-0.js	jasmine-cleanup-0	446 bytes
Initial total		3.82 MB

```
Application bundle generation complete. [4.450 seconds] - 2026-10-05T04:08:26.751Z
```

```
Watch mode enabled. Watching for file changes...
```

```
04 10 2026 23:08:27.136:WARN [karma]: No captured browser, open http://localhost:9876/
```

```
04 10 2026 23:08:27.191:INFO [karma-server]: Karma v6.4.4 server started at http://localhost:9876/
```

```
04 10 2026 23:08:27.192:INFO [launcher]: Launching browsers Chrome with concurrency unlimited
```

```
04 10 2026 23:08:27.213:INFO [launcher]: Starting browser Chrome
```

```
04 10 2026 23:08:28.961:INFO [Chrome 154.0.0.0 (Windows 10)]: Connected on socket dN7GY_qSUEUfB6owAAAA with id 46638089
```

```
Chrome 154.0.0.0 (Windows 10): Executed 4 of 4 SUCCESS (0.195 secs / 0.172 secs)
```

```
TOTAL: 4 SUCCESS
```